BEHAVIORAL DESIGN DOCUMENT

Use Case Diagrams

Complete Actor-System Interaction Reference — All 5 Roles & 6 Bounded Contexts

Version: 2.0 | **Date:** April 2026

Actors: Dealer, Admin, Warehouse, Logistics, Delivery Agent

System Actors: Outbox Dispatcher, Notification Consumer, Email Dispatcher

Bounded Contexts: Identity, Catalog, Order, Logistics, Payment, Notification

Classification: Confidential — Internal Use Only

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1. Actor Inventory & Role Matrix

1.1 Human Actors

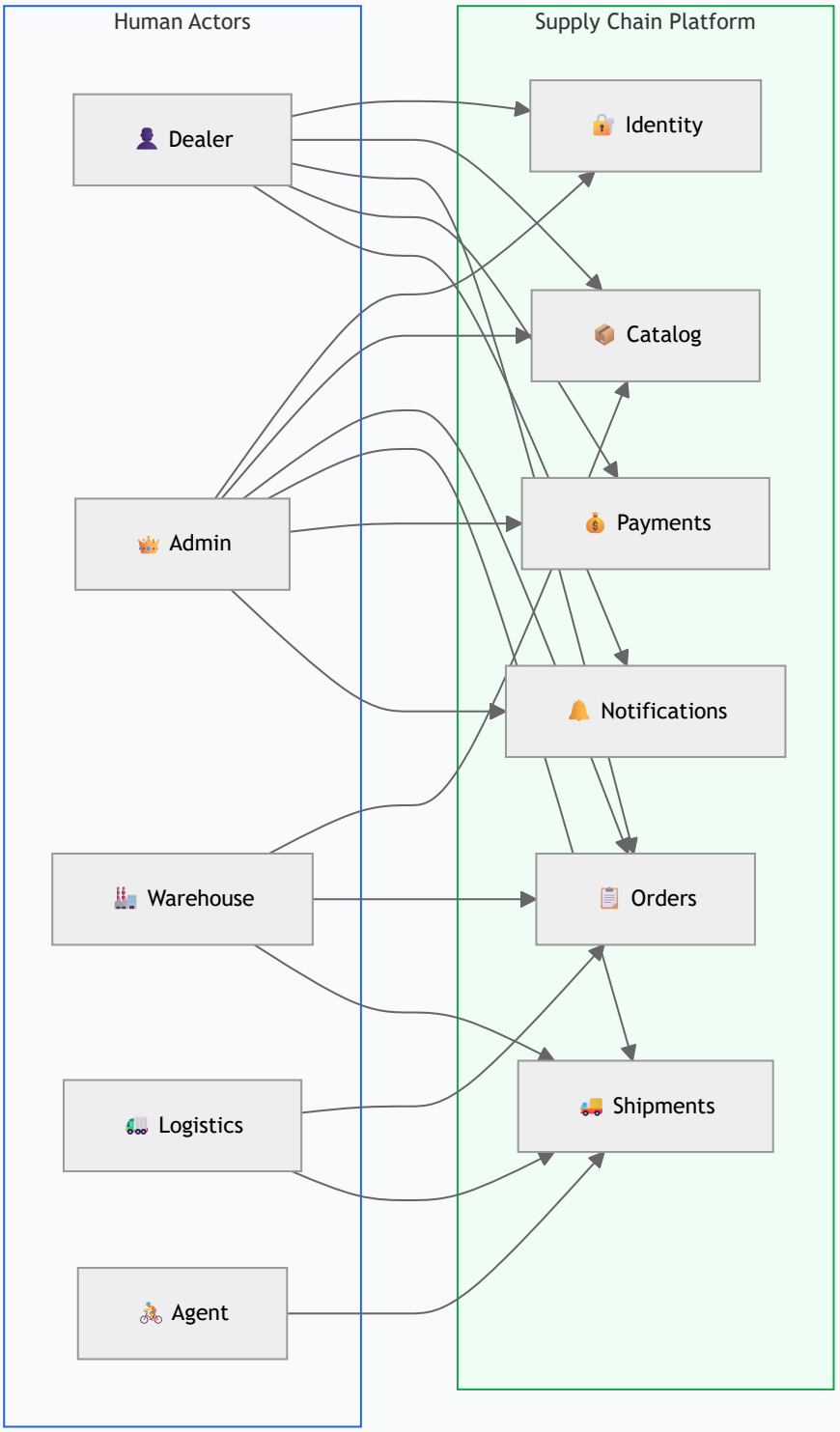

FIGURE 1 — ACTOR-TO-MODULE ACCESS MAP

1.2 Role Permission Matrix

ACTOR	DESCRIPTION	AUTH LEVEL	ACCESSIBLE MODULES
Dealer	B2B customer registered via self-service	JWT (Approved status)	Catalog, Orders, Cart, Invoices, Shipments (own), Notifications, Profile
Admin	Platform operator with full system access	JWT (Admin role)	All modules — full CRUD + dealer management + analytics
Warehouse	Fulfillment team — stock and dispatch	JWT (Warehouse role)	Catalog (restock), Orders (status), Shipments (create)
Logistics	Logistics coordinator — shipment routing	JWT (Logistics role)	Orders (status), Shipments (assign agent, status), AI chatbot
Agent	Last-mile delivery agent	JWT (Agent role)	Shipments (assigned, status update, accept/reject)

1.3 System Actors

ACTOR	TYPE	PURPOSE
Outbox Dispatcher	Background Worker	Polls outbox tables and publishes events to RabbitMQ
Notification Consumer	RabbitMQ Consumer	Consumes domain events and creates notification records
Email Dispatcher	Background Worker	Resolves recipient emails and sends via SMTP
RabbitMQ	Message Broker	Topic exchange routing for all domain events
Redis	Cache Store	Token revocation and command idempotency

2. System-Level Use Case Overview

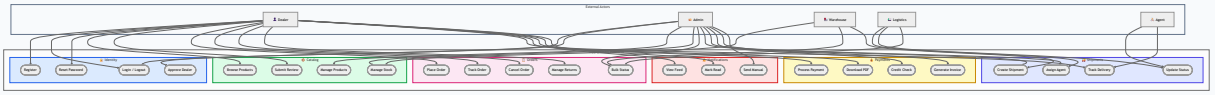

FIGURE 2 — SYSTEM-LEVEL USE CASE OVERVIEW (ALL ACTORS AND BOUNDED CONTEXTS)

3. UC-1: Identity & Authentication

IdentityAuth Service

8 Use Cases · Port 8001

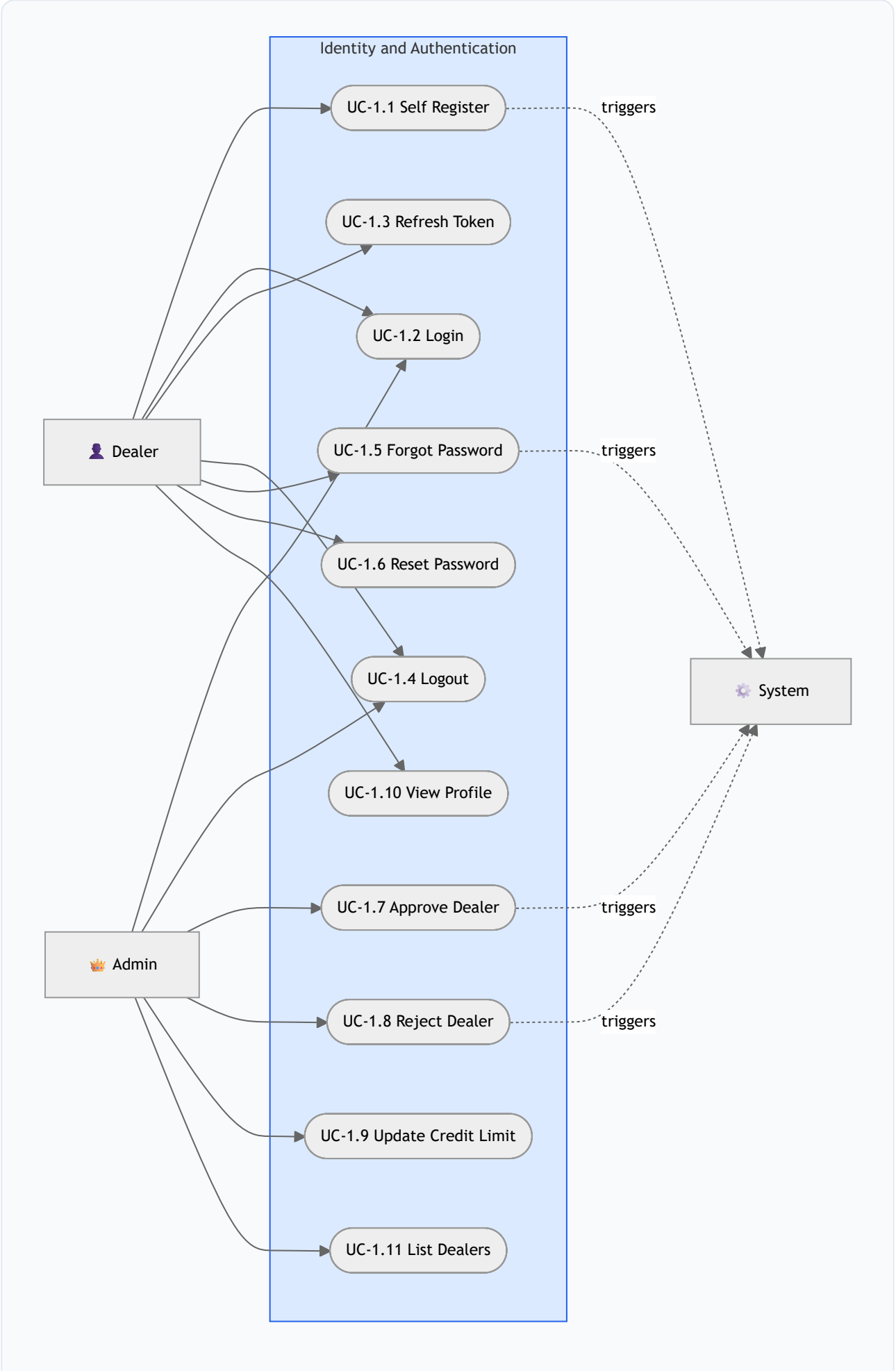
Identity and Authentication
UC-1.1 Self Register
UC-1.3 Refresh Token
UC-1.2 Login
UC-1.5 Forgot Password
UC-1.6 Reset Password
UC-1.4 Logout
UC-1.10 View Profile
UC-1.7 Approve Dealer
UC-1.8 Reject Dealer
UC-1.9 Update Credit Limit
UC-1.11 List Dealers
Dealer
Admin
System
triggers
triggers
triggers
triggers

FIGURE 3 — UC-1: IDENTITY AND AUTHENTICATION USE CASES

3.1 Use Case Specifications

ID	USE CASE	ACTOR	PRECONDITION	PRIMARY FLOW	EVENTS EMITTED
UC-1.1	Self Register	Dealer	Not registered	Submit form with business details and GST → create User (Pending) + DealerProfile → publish event	DealerRegistered
UC-1.2	Login	Any	Approved status	Email + Password → validate → issue JWT + Refresh Token cookie	—
UC-1.3	Refresh Token	Any	Valid refresh cookie	Validate refresh token → issue new JWT + rotate refresh	—
UC-1.4	Logout	Any	Authenticated	Revoke refresh token → add JTI to Redis revocation set	—
UC-1.5	Forgot Password	Dealer	Registered	Email → generate OTP → hash and store → trigger email	PasswordResetRequested
UC-1.6	Reset Password	Dealer	Valid OTP	OTP + new password → verify OTP → update hash → mark OTP used	PasswordResetCompleted
UC-1.7	Approve Dealer	Admin	Dealer in Pending	Set status = Approved → create credit account in PaymentService → publish event	DealerApproved
UC-1.8	Reject Dealer	Admin	Dealer in Pending	Set status = Rejected with reason → publish event	DealerRejected
UC-1.9	Update Credit Limit	Admin	Dealer approved	Update CreditLimit field → call PaymentService internal API	CreditLimitUpdated
UC-1.10	View Profile	Any	Authenticated	Return current user + dealer profile if applicable	—
UC-1.11	List Dealers	Admin	Admin role	Paginated dealer list with status and role filters	—

4. UC-2: Catalog & Inventory Management

CatalogInventory Service

12 Use Cases · Port 8002

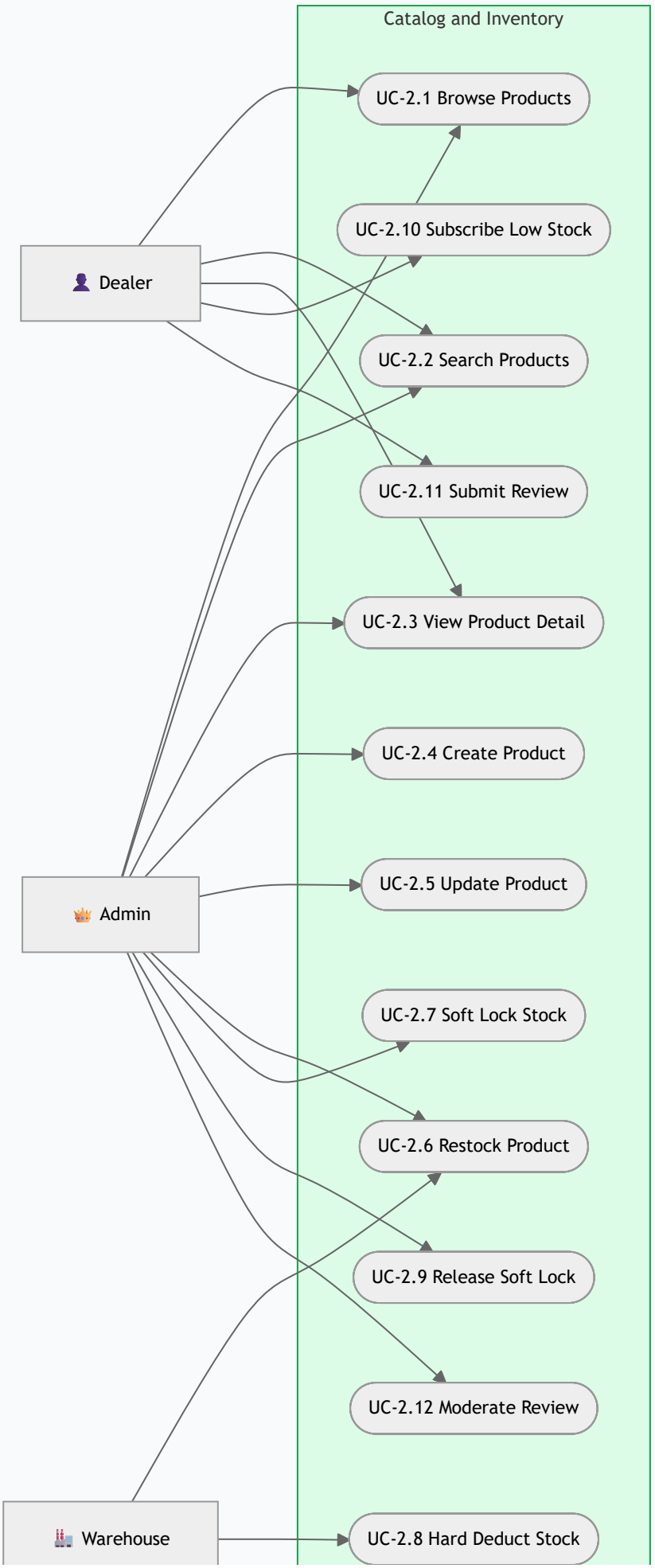
Catalog and Inventory
UC-2.1 Browse Products
UC-2.10 Subscribe Low Stock
Dealer
UC-2.2 Search Products
UC-2.11 Submit Review
UC-2.3 View Product Detail
UC-2.4 Create Product
Admin
UC-2.5 Update Product
UC-2.7 Soft Lock Stock
UC-2.6 Restock Product
UC-2.9 Release Soft Lock
UC-2.12 Moderate Review
Warehouse
UC-2.8 Hard Deduct Stock

FIGURE 4 — UC-2: CATALOG AND INVENTORY USE CASES

4.1 Use Case Specifications

ID	USE CASE	ACTOR	PRECONDITION	PRIMARY FLOW	EVENTS EMITTED
UC-2.1	Browse Products	Any	None	Paginated product list with category filter and sorting	—
UC-2.2	Search Products	Any	None	Keyword search across name / SKU / description	—
UC-2.3	View Product Detail	Any	None	Full product info + stock availability + reviews	—
UC-2.4	Create Product	Admin	Admin role	Product name + SKU + price + category → create with initial stock	ProductCreated
UC-2.5	Update Product	Admin	Product exists	Update name / price / description / active status	ProductUpdated
UC-2.6	Restock Product	Admin / Warehouse	Product exists	Add quantity → create StockTransaction(Restock) → increase PhysicalStock	StockUpdated
UC-2.7	Soft Lock Stock	System	Order placed	Reserve quantity → create StockTransaction(SoftLock) → increase LockedStock	—
UC-2.8	Hard Deduct Stock	System	Shipment dispatched	Deduct from physical → create StockTransaction(HardDeduct) → decrease both	StockUpdated
UC-2.9	Release Soft Lock	System	Order cancelled	Reverse lock → create StockTransaction(Release) → decrease LockedStock	—
UC-2.10	Subscribe Low Stock	Dealer	Authenticated	Set alert threshold for a product → notify when stock drops below	—
UC-2.11	Submit Review	Dealer	Has purchased product	Submit rating + comment → pending moderation	—
UC-2.12	Moderate Review	Admin	Review pending	Approve or reject review → update visibility	—

5. UC-3: Order Lifecycle

Order Service

11 Use Cases · Port 8003

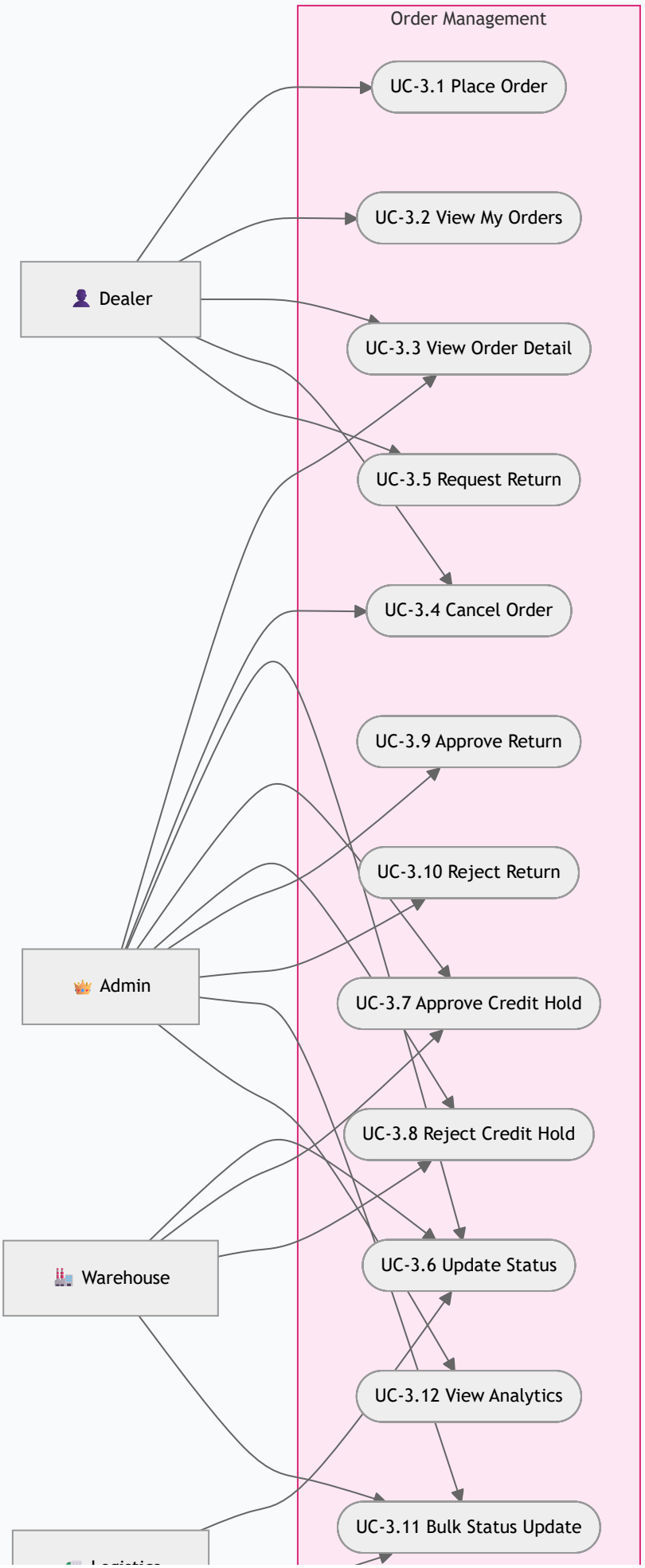
Order Management
UC-3.1 Place Order
UC-3.2 View My Orders
Dealer
UC-3.3 View Order Detail
UC-3.5 Request Return
UC-3.4 Cancel Order
UC-3.9 Approve Return
UC-3.10 Reject Return
Admin
UC-3.7 Approve Credit Hold
UC-3.8 Reject Credit Hold
Warehouse
UC-3.6 Update Status
UC-3.12 View Analytics
UC-3.11 Bulk Status Update

FIGURE 5 — UC-3: ORDER LIFECYCLE USE CASES

5.1 Use Case Specifications

ID	USE CASE	ACTOR	PRECONDITION	PRIMARY FLOW	EVENTS EMITTED
UC-3.1	Place Order	Dealer	Cart has items	Submit cart → credit check → soft lock stock → if approved: Processing / if rejected: OnHold	OrderPlaced or CreditHoldRequired
UC-3.2	View My Orders	Dealer	Authenticated	Paginated list of dealer's own orders with status filter	—
UC-3.3	View Order Detail	Any	Ownership check	Order aggregate + lines + status history + return request + saga state	—
UC-3.4	Cancel Order	Dealer / Admin	Cancellable status	Validate state machine transition → release stock lock → set Cancelled	OrderCancelled
UC-3.5	Request Return	Dealer	Order = Delivered	Submit reason → create ReturnRequest → set ReturnRequested	ReturnRequested
UC-3.6	Update Status	Admin / WH / Log	Valid transition	Validate against state machine → record in StatusHistory → emit event	OrderStatusUpdated
UC-3.7	Approve Credit Hold	Admin / WH	Order = OnHold	Set CreditHoldStatus = Approved → status = Processing → soft lock stock	OrderPlaced
UC-3.8	Reject Credit Hold	Admin / WH	Order = OnHold	Set CreditHoldStatus = Rejected → status = Cancelled	OrderCancelled
UC-3.9	Approve Return	Admin	ReturnRequested	ReturnRequest.IsApproved = true → status = ReturnApproved → close	ReturnApproved
UC-3.10	Reject Return	Admin	ReturnRequested	ReturnRequest.IsApproved = false → status = ReturnRejected → close	ReturnRejected
UC-3.11	Bulk Status Update	Admin / WH / Log	Valid orders	Select multiple orders → validate all transitions → apply or preview	Multiple OrderStatusUpdated
UC-3.12	View Analytics	Admin	Admin role	Purchase KPIs and top dealers/products aggregation	—

6. UC-4: Logistics & Shipment Tracking

LogisticsTracking Service

9 Use Cases · Port 8004

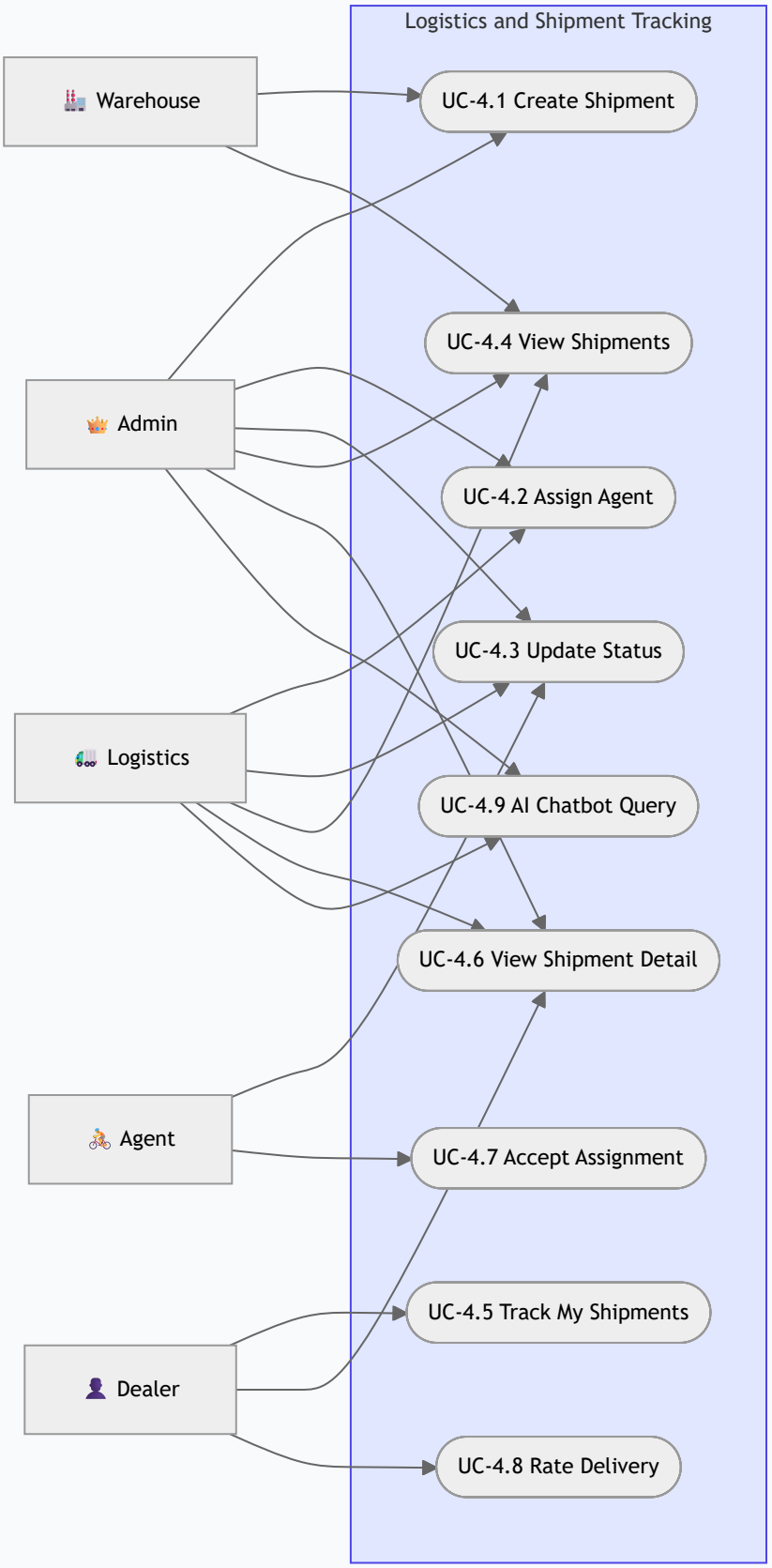

FIGURE 6 — UC-4: LOGISTICS AND SHIPMENT TRACKING USE CASES

6.1 Use Case Specifications

ID	USE CASE	ACTOR	PRECONDITION	PRIMARY FLOW	EVENTS EMITTED
UC-4.1	Create Shipment	Admin / WH	Order = ReadyForDispatch	Create shipment with delivery address → status = Created → link to order	ShipmentCreated
UC-4.2	Assign Agent	Admin / Logistics	Shipment = Created	Select agent + vehicle → status = Assigned → emit event	ShipmentAssigned
UC-4.3	Update Status	Admin / Log / Agent	Valid transition	Validate shipment state machine → create ShipmentEvent → emit	ShipmentStatusUpdated
UC-4.4	View Shipments	Admin / WH / Log	Authenticated	Paginated shipment list with status and date filters	—
UC-4.5	Track My Shipments	Dealer	Authenticated	View own shipments filtered by DealerId with event timeline	—
UC-4.6	View Shipment Detail	Any	Ownership/role check	Full shipment info + event timeline + ops state	—
UC-4.7	Accept Assignment	Agent	Assigned to agent	Agent accepts/rejects assignment → update decision status	ShipmentStatusUpdated
UC-4.8	Rate Delivery	Dealer	Shipment = Delivered	Submit 1-5 rating + optional comment → persist on shipment	—
UC-4.9	AI Chatbot Query	Admin / Log / Agent	Authenticated	Natural language query → AI processes shipment context → return insights	—

7. UC-5: Payment & Invoicing

PaymentInvoice Service

10 Use Cases · Port 8005

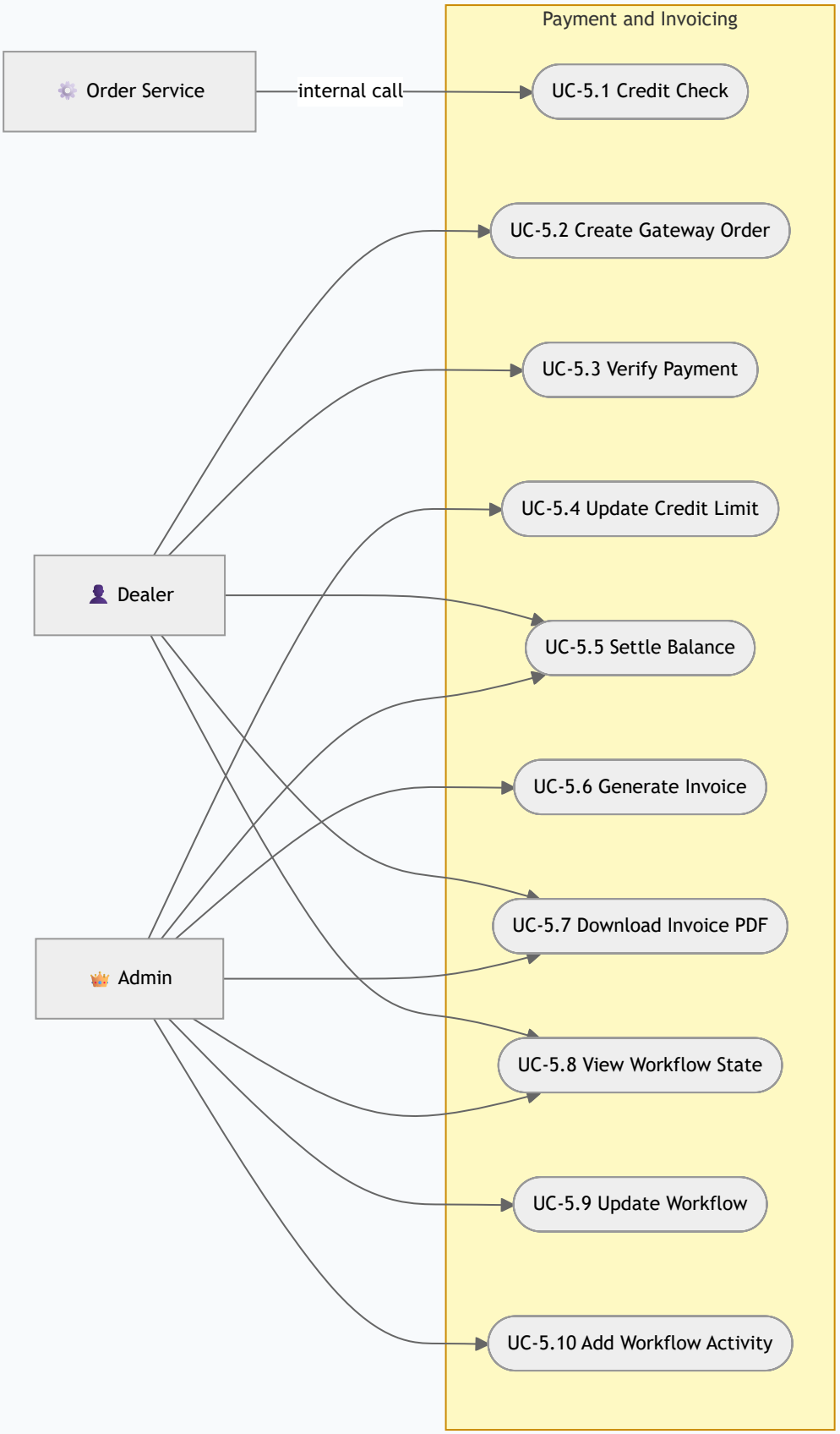

FIGURE 7 — UC-5: PAYMENT AND INVOICING USE CASES

7.1 Use Case Specifications

ID	USE CASE	ACTOR	PRECONDITION	PRIMARY FLOW	EVENTS EMITTED
UC-5.1	Credit Check	System	Order being placed	Lookup DealerCreditAccount → compare AvailableCredit vs order amount → return pass/fail	—
UC-5.2	Create Gateway Order	Dealer	Order confirmed	Create Razorpay order → return order ID and key to frontend	—
UC-5.3	Verify Payment	Dealer	Razorpay checkout done	HMAC-SHA256 signature verify → record payment → optionally capture	PaymentVerified
UC-5.4	Update Credit Limit	Admin	Dealer approved	Update CreditLimit on DealerCreditAccount	CreditLimitUpdated
UC-5.5	Settle Balance	Admin / Dealer	Outstanding > 0	Apply settlement amount → reduce CurrentOutstanding	—
UC-5.6	Generate Invoice	Admin	Order exists	Idempotency check → build lines from order → calculate GST → generate QuestPDF → persist	InvoiceGenerated
UC-5.7	Download Invoice PDF	Admin / Dealer	Invoice exists	Return stored PDF file for download	—
UC-5.8	View Workflow State	Admin / Dealer	Invoice exists	Return current collection workflow state and due date	—
UC-5.9	Update Workflow	Admin	Workflow exists	Upsert workflow state (Pending → Reminded → Overdue → Collected)	—
UC-5.10	Add Workflow Activity	Admin / Dealer	Invoice exists	Append activity log entry (Created / Reminded / Escalated / Collected / Note)	—

8. UC-6: Notification Engine

Notification Service
5 Use Cases · Port 8006

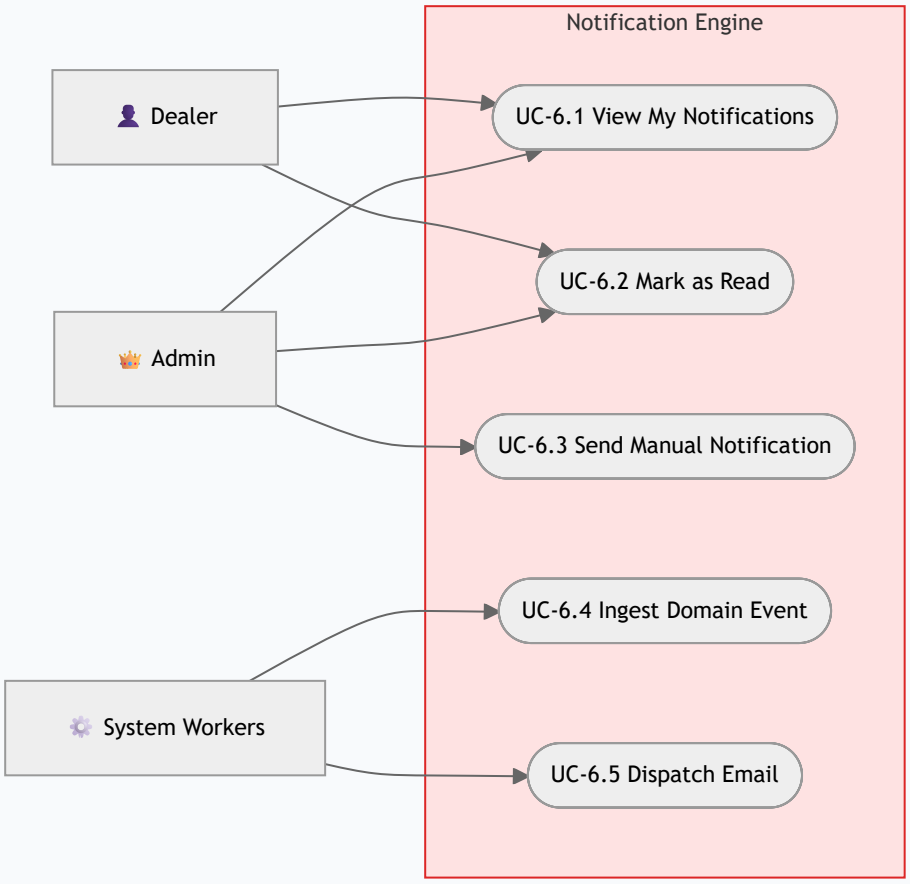

FIGURE 8 — UC-6: NOTIFICATION ENGINE USE CASES

8.1 Use Case Specifications

ID	USE CASE	ACTOR	PRECONDITION	PRIMARY FLOW	EVENTS EMITTED
UC-6.1	View My Notifications	Any	Authenticated	Paginated feed filtered by RecipientUserId. Admin sees all.	—
UC-6.2	Mark as Read	Any	Owns notification	Set IsRead = true and ReadAtUtc = now	—
UC-6.3	Send Manual Notification	Admin	Admin role	Create notification with title + body + channel + recipient	—
UC-6.4	Ingest Domain Event	System	Event published	Consumer receives from RabbitMQ → classify channel → create notification record	—
UC-6.5	Dispatch Email	System	Email-channel pending	Poll pending → resolve email via IdentityAuth → send HTML via SMTP → mark sent/failed	—

9. UC-7: AI Exception Playbooks

AI Decision Support
4 Use Cases · Human-in-the-Loop

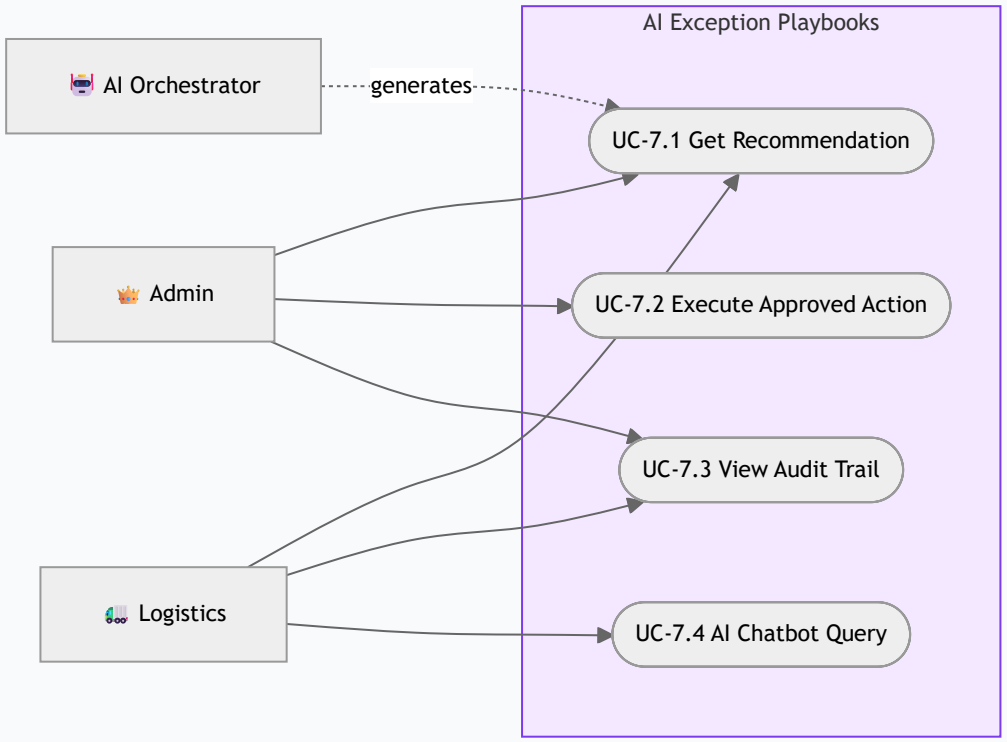

FIGURE 9 — UC-7: AI EXCEPTION PLAYBOOKS USE CASES

9.1 Use Case Specifications

ID	USE CASE	ACTOR	PRECONDITION	PRIMARY FLOW	GOVERNANCE
UC-7.1	Get Recommendation	Admin / Logistics	Exception detected	Gather entity context → AI generates recommendation → validate schema → return with confidence score	Confidence threshold + schema validation
UC-7.2	Execute Approved Action	Admin	Recommendation approved	Execute suggested actions in sequence → record outcome → update audit trail	Human approval required for medium/high impact
UC-7.3	View Audit Trail	Admin / Logistics	Exception exists	View full history: exception → recommendation → decision → outcome	Immutable audit log
UC-7.4	AI Chatbot Query	Admin / Logistics	Authenticated	Natural language question about shipments → AI responds with insights and suggested actions	Read-only — no state mutations

10. Cross-Cutting System Automation

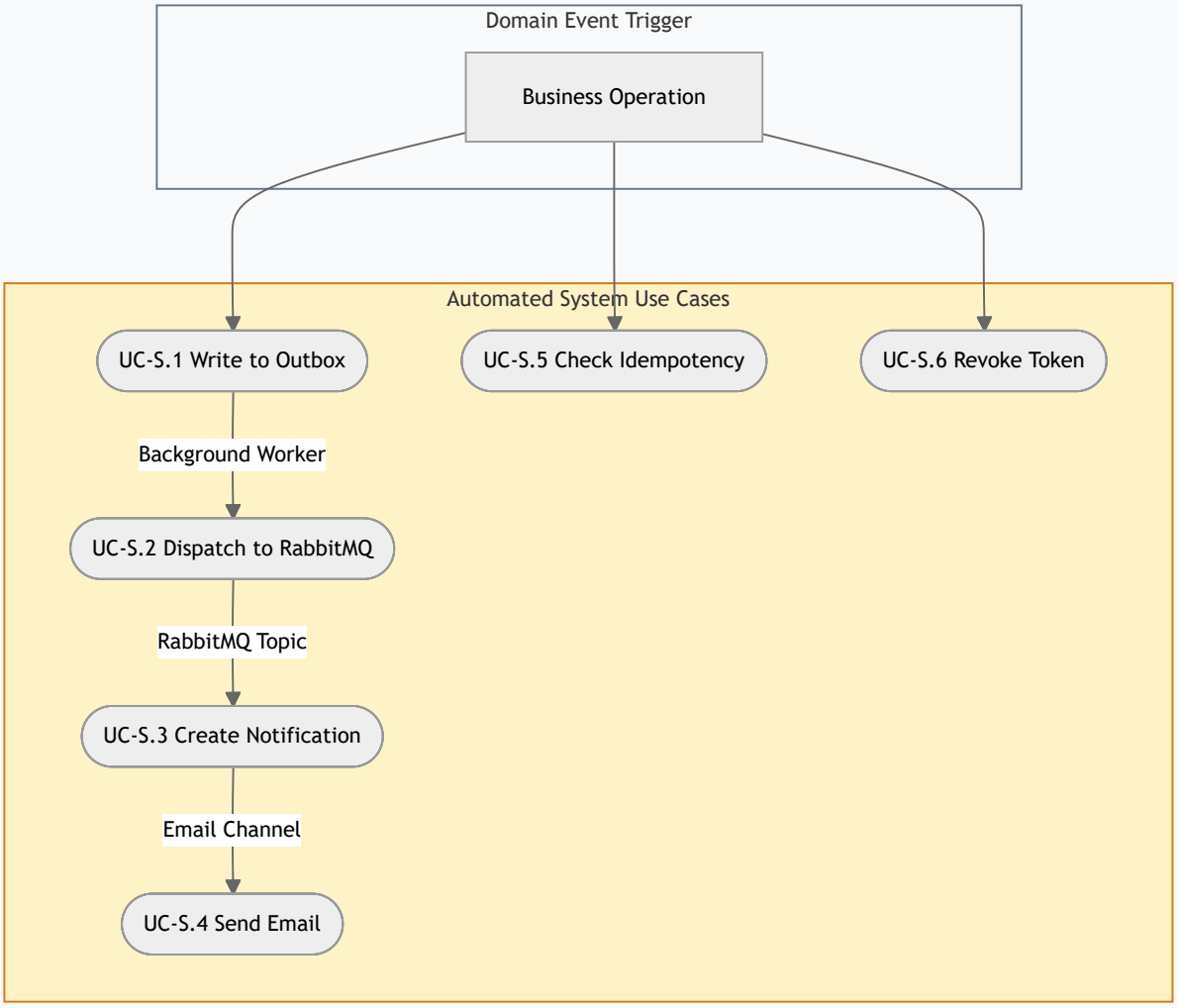

FIGURE 10 — CROSS-CUTTING SYSTEM AUTOMATION FLOW

ID	SYSTEM USE CASE	TRIGGER	FLOW	FAILURE HANDLING
UC-S.1	Write to Outbox	Any domain write	Domain data + OutboxMessage in single transaction	Fail = no event (strong consistency)
UC-S.2	Dispatch to RabbitMQ	Background poll	Read Pending messages → publish → mark Published	Retry up to max → mark Failed
UC-S.3	Create Notification	RabbitMQ event	Consumer → classify channel → persist notification	Dead-letter queue on repeated failure
UC-S.4	Send Email	Email-channel pending	Resolve email via Identity → send HTML via SMTP	Mark failed + retry next cycle
UC-S.5	Check Idempotency	Any POST/PUT command	Check Redis for key → skip if exists → set key with TTL	Redis unavailable → allow through
UC-S.6	Revoke Token	Logout	Add JTI to Redis revocation set with token TTL	Redis unavailable → log warning

11. Use Case Traceability Matrix

Maps each use case to its implementing API endpoint and service:

USE CASE ID	NAME	SERVICE	API ENDPOINT	HTTP METHOD
UC-1.1	Self Register	IdentityAuth	/api/auth/register	POST
UC-1.2	Login	IdentityAuth	/api/auth/login	POST
UC-1.5	Forgot Password	IdentityAuth	/api/auth/forgot-password	POST
UC-1.7	Approve Dealer	IdentityAuth	/api/admin/dealers/{id}/approve	PUT
UC-2.1	Browse Products	CatalogInventory	/api/products	GET
UC-2.4	Create Product	CatalogInventory	/api/products	POST
UC-2.6	Restock Product	CatalogInventory	/api/products/{id}/restock	POST
UC-2.7	Soft Lock Stock	CatalogInventory	/api/inventory/soft-lock	POST
UC-2.11	Submit Review	CatalogInventory	/api/products/{id}/reviews	POST
UC-3.1	Place Order	Order	/api/orders	POST
UC-3.4	Cancel Order	Order	/api/orders/{id}/cancel	POST
UC-3.5	Request Return	Order	/api/orders/{id}/returns	POST
UC-3.7	Approve Credit Hold	Order	/api/admin/orders/{id}/approve-hold	PUT
UC-3.11	Bulk Status Update	Order	/api/admin/orders/bulk-status	POST
UC-3.12	View Analytics	Order	/api/admin/orders/analytics	GET
UC-4.1	Create Shipment	Logistics	/api/logistics/shipments	POST
UC-4.2	Assign Agent	Logistics	/api/logistics/shipments/{id}/assign-agent	PUT
UC-4.9	AI Chatbot Query	Logistics	/api/logistics/shipments/chatbot/ask	POST
UC-5.1	Credit Check	PaymentInvoice	/api/payment/dealers/{id}/credit-check	GET
UC-5.6	Generate Invoice	PaymentInvoice	/api/payment/invoices	POST
UC-5.7	Download Invoice	PaymentInvoice	/api/payment/invoices/{id}/download	GET
UC-6.1	View Notifications	Notification	/api/notifications/my	GET
UC-6.3	Send Manual	Notification	/api/notifications/manual	POST
UC-7.1	Get Recommendation	AI Orchestrator	/api/ai/exceptions/{id}/recommend	POST
UC-7.2	Execute Approved	AI Orchestrator	/api/ai/exceptions/{id}/execute	POST

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